

Analytical solutions for frequency estimators by interpolation of DFT coefficients



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ABSTRACT

For the estimation of the frequency of a complex sinusoid, previous methods interpolate DFT coefficients to obtain the frequency offset. Most of these interpolation methods can be considered as linear approximation of a nonlinear function. Up to date, only a few previously proposed estimators have been shown to have analytical expressions. In this paper, we show that almost all the interpolated estimators proposed to date, both direct and iterative, have simple analytical solutions for the frequency offset. Thus, the estimation bias of these estimators can be better understood and a whole new set of unbiased estimators using the analytical solutions is also introduced.

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1. Introduction

Frequency estimation of a complex sinusoidal signal is a fundamental problem in digital signal processing and has applications in areas such as biomedical signal processing, radar analysis, signal detection and telecommunications [1]. The signal can be described as

$$s[n] = A_0 e^{j(2\pi(f_0/f_s)n + \theta_0)} + w[n], \quad n = 0, 1, \dots, N-1 \quad (1)$$

where A_0 , f_0 , and θ_0 are the amplitude, frequency, and phase of the signal, respectively, the term $w[n]$ is an additive noise, f_s is the sampling frequency, and N is the number of samples.

The ML estimator for the frequency estimation under additive white Gaussian observation noise is the peak search of the periodogram corresponding to the input

signal $s[n]$ [2]:

$$\hat{f}_{0,ML} = \arg \max_f \left\{ \left| \sum_{n=0}^{N-1} s[n] e^{-j2\pi n f} \right| \right\}. \quad (2)$$

The Cramér Rao lower bound (CRLB) of the mean squared error (MSE) is

$$\sigma_f^2 = \frac{6f_s^2}{4\pi^2 N(N^2 - 1)\rho} \quad (3)$$

where ρ is the SNR of $s[n]$.

The search for the frequency is usually divided into two steps: a coarse search and a fine search. The coarse search finds the index of the peak magnitude, k_p , of the discrete Fourier transform (DFT):

$$S[k] = \sum_{n=0}^{N-1} s[n] e^{-j(2\pi/N)nk}. \quad (4)$$

$$k_p = \arg \max_k |S[k]|. \quad (5)$$

The fine search finds a frequency offset $\hat{\delta}$ in the vicinity of k_p with the constraint $|\hat{\delta}| < 1/2$. The estimated frequency is

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$$\hat{f}_0 = (k_p + \hat{\delta}) \frac{f_s}{N}. \quad (6)$$

In the fine search, $\hat{\delta}$ is calculated by interpolating the DFT coefficients in the neighborhood of $S[k_p]$ [3–11].

Since the exact relationship between the frequency offset and the DFT coefficient interpolation is nonlinear, most of these direct interpolation approaches are inevitably approximation methods by linearizing the nonlinear function and cause estimation bias. In this case, the MSE has two components [12]:

$$\text{MSE} = \sigma_e^2 + \text{bias}^2. \quad (7)$$

The first component, σ_e^2 , is contributed by the additive noise $w[n]$ in (1) and its lower bound is characterized by CRLB. The second component, bias, is caused by the linearization of the interpolation. From the composition of MSE, we can see that the performance of an estimator can be divided into two regions according to the noise level. At low SNR ($\sigma_e \gg \text{bias}$), σ_e is the dominant term and the performance of an estimator is determined by SNR. On the other hand, at high SNR ($\sigma_e \ll \text{bias}$), the bias term is the determining factor and the performance of an estimator is limited by the bias. A general discussion on these two limiting factors can be found in [12]. To make the estimation more accurate, either σ_e or bias has to be reduced. In this regard, many iterative approaches [2,6,13–22] have been proposed to refine the solution. As the number of iteration increases, the bias can either be neglected or eliminated and Aboutanios and Mulgrew [17] have shown that σ_e^2 of their iterative estimator can approach CRLB asymptotically in two iterations. The bias can also be reduced by zero padding [23,24] or solving its nonlinear function [23–25]. However, the most efficient way to completely eliminate the bias is to remove the approximation from the estimation.

Therefore, to obtain a truly unbiased estimator, nonlinear analytical solution is necessary. In this regard, three analytical solutions have been proposed to date. The first was proposed by Bertocco et al. [26–29] using two neighboring DFT coefficients. The second was derived by Aboutanios [27] from their iterative estimator. The third was proposed by Candan [12] during submission of this paper. It shows that Jacobsen estimator has an analytical solution. In this paper, we will show that analytical solutions are not limited to these two cases and almost all the interpolated estimators proposed to date have simple analytical solutions for the frequency offset $\hat{\delta}$. These analytical expressions have both practical and theoretical values. Practically, these expressions allow us to have estimators that are accurate without iteration and are only sensitive to noise. Thus, a whole new set of unbiased estimators is introduced in the paper. Theoretically, the solutions provide a basis to understand the estimation bias caused by the approximation. In the experiments, we show that these analytical solutions do indeed provide accurate estimation without bias when noise is not present.

The remainder of the paper is organized as follows. In the second section, we present the theoretical analysis that leads to the analytical solutions of the estimators and discusses possible variations of the analytical estimators.

In the third section, we show the experimental results. Finally, in the fourth section, we conclude.

2. Theoretical analysis of the estimators

The first step of the analysis is to derive the analytical expression for the five spectrum lines that will be used in the interpolation: $S[k_p - 1]$, $S[k_p]$, $S[k_p + 1]$, $S[k_p - 1/2]$, and $S[k_p + 1/2]$. Plugging (1) and (6) into (4) and set the noise $w[n]$ to zero, we get

$$S[k] = A_0 e^{j\theta_0} \sum_{n=0}^{N-1} e^{-j2\pi((k-k_p-\delta)/N)n}. \quad (8)$$

It is the sum of a geometric series and the five spectrum lines can be expressed as the following general equation:

$$S[k_p + k_d] = A_0 e^{j\theta_0} e^{j((N-1)\pi(\delta-k_d)/N)} \frac{\sin(\pi(\delta-k_d))}{\sin\left(\frac{\pi(\delta-k_d)}{N}\right)}. \quad (9)$$

where $k_d = 0, \pm 1, \pm 1/2$. With this equation, we can now analyze the interpolated estimators.

2.1. Bertocco estimator

First, let us look at the analytical solution proposed by Bertocco et al. [26] and further explored by Aboutanios [27,28]. It can be written as follows:

$$\hat{\delta}_{ber} = \frac{N}{2\pi} \arg \left\{ \frac{S[k_p] - S[k_p \pm 1]}{S[k_p] - e^{\pm(-j2\pi/N)} S[k_p \pm 1]} \right\} \quad (10)$$

where positive sign will be used if $S[k_p + 1] \geq S[k_p - 1]$ and vice versa. This equation is slightly different than the original equation but they are mathematically equivalent. Considering the case of positive sign and applying (9), we can achieve a derivation which is different from Bertocco's but follows the same methodology throughout the paper:

$$\begin{aligned} & \frac{S[k_p] - S[k_p + 1]}{S[k_p] - e^{-j(2\pi/N)} S[k_p + 1]} \\ &= \frac{\frac{1}{\sin(\pi\delta/N)} - e^{j(\pi/N)} \frac{1}{\sin(\pi(\delta-1)/N)}}{\frac{1}{\sin(\pi\delta/N)} - e^{-j(\pi/N)} \frac{1}{\sin(\pi(\delta-1)/N)}} \\ &= \frac{\sin\left(\frac{\pi(\delta-1)}{N}\right) - \cos\left(\frac{\pi}{N}\right) \sin\left(\frac{\pi\delta}{N}\right) - j \sin\left(\frac{\pi}{N}\right) \sin\left(\frac{\pi\delta}{N}\right)}{\sin\left(\frac{\pi(\delta-1)}{N}\right) - \cos\left(\frac{\pi}{N}\right) \sin\left(\frac{\pi\delta}{N}\right) + j \sin\left(\frac{\pi}{N}\right) \sin\left(\frac{\pi\delta}{N}\right)} \\ &= \frac{-\sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) - j \sin\left(\frac{\pi}{N}\right) \sin\left(\frac{\pi\delta}{N}\right)}{-\sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) + j \sin\left(\frac{\pi}{N}\right) \sin\left(\frac{\pi\delta}{N}\right)} = e^{j(2\pi\delta/N)}. \end{aligned} \quad (11)$$

For the above and the following derivations, we use the relationship $\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$ when $\sin(\pi(\delta \pm 1)/N)$ is involved.

Aboutanios [27] has further studied the more general case of Bertocco estimator for damped cisoid signals. By removing the damping factor, the simplified result from [27] is that (11) can be approximated as $1 + j2\pi\delta/N$ and the frequency offset can be approximated by the imaginary part.

Therefore, we can express (10) in an alternative form as

$$\hat{\delta}_{ber} = \frac{N}{2\pi} \sin^{-1} \left\{ \operatorname{Im} \left\{ \frac{S[k_p] - S[k_p \pm 1]}{S[k_p] - e^{\pm (-j2\pi/N)S[k_p \pm 1]}} \right\} \right\}. \quad (12)$$

Although this alternative equation probably does not have practical values, it is useful to relate Bertocco estimator to Quinn estimator which will be discussed next.

Examining (11), a slight variation which has not yet been proposed anywhere can be written as follows:

$$\hat{\delta}'_{ber} = \frac{N}{\pi} \arg \left\{ \frac{S[k_p] - S[k_p \pm 1]}{S[k_p] - e^{\pm (-j\pi/N)S[k_p \pm 1]}} \right\}. \quad (13)$$

The above equation essentially makes the denominator a real number and extracts the phase from the numerator. In contrast, (11) makes the phases of the numerator and denominator opposite. We will call this variation as version 2 of the Bertocco estimator.

2.2. Quinn estimator

The estimator proposed by Quinn [3,4] works as follows:

$$\hat{\delta}_{qui1} = \frac{\operatorname{Re} \left\{ \frac{S[k_p - 1]}{S[k_p]} \right\}}{1 - \operatorname{Re} \left\{ \frac{S[k_p - 1]}{S[k_p]} \right\}}, \quad (14)$$

$$\hat{\delta}_{qui2} = \frac{\operatorname{Re} \left\{ \frac{S[k_p + 1]}{S[k_p]} \right\}}{\operatorname{Re} \left\{ \frac{S[k_p + 1]}{S[k_p]} \right\} - 1}. \quad (15)$$

If $\hat{\delta}_{qui1} > 0$ and $\hat{\delta}_{qui2} > 0$, the final estimation is $\hat{\delta}_{qui} = \hat{\delta}_{qui2}$. Otherwise, $\hat{\delta}_{qui} = \hat{\delta}_{qui1}$. Closer inspection on the above criteria can lead to a simpler implementation, i.e., if $\hat{\delta}_{qui1} < 0$, we can apply $\hat{\delta}_{qui} = \hat{\delta}_{qui1}$ and only $\hat{\delta}_{qui1}$ needs to be calculated. Otherwise, both equations need to be calculated. On the other hand, we can also use the criteria in Bertocco's estimator, i.e., if $S[k_p - 1] > S[k_p + 1]$, $\hat{\delta}_{qui} = \hat{\delta}_{qui2}$ and vice versa. Then, only one equation needs to be calculated. However, when noise is large, using Bertocco's rule may be less robust especially when δ is close to 0 [3]. In this case, despite the possibility of needing to calculate two equations, Quinn's rule has the advantage because it uses all three spectrum lines ($S[k_p - 1]$, $S[k_p]$, and $S[k_p + 1]$) instead of just two as in Bertocco's rule.

To show that the above two equations have analytical solutions for δ , we can apply (9) to obtain the following two equations:

$$\begin{aligned} \operatorname{Re} \left\{ \frac{S[k_p - 1]}{S[k_p]} \right\} &= \operatorname{Re} \left\{ e^{-j(\pi/N)} \frac{\sin \left(\frac{\pi\delta}{N} \right)}{\sin \left(\frac{\pi(\delta+1)}{N} \right)} \right\} \\ &= \frac{\cos \left(\frac{\pi}{N} \right) \sin \left(\frac{\pi\delta}{N} \right)}{\sin \left(\frac{\pi(\delta+1)}{N} \right)}, \end{aligned} \quad (16)$$

$$\operatorname{Re} \left\{ \frac{S[k_p + 1]}{S[k_p]} \right\} = \frac{\cos \left(\frac{\pi}{N} \right) \sin \left(\frac{\pi\delta}{N} \right)}{\sin \left(\frac{\pi(\delta-1)}{N} \right)}. \quad (17)$$

Then, we can obtain

$$\begin{aligned} \hat{\delta}_{qui1} &= \frac{\cos \left(\frac{\pi}{N} \right) \sin \left(\frac{\pi\delta}{N} \right)}{\sin \left(\frac{\pi(\delta+1)}{N} \right) - \cos \left(\frac{\pi}{N} \right) \sin \left(\frac{\pi\delta}{N} \right)} \\ &= \frac{\cos \left(\frac{\pi}{N} \right) \sin \left(\frac{\pi\delta}{N} \right)}{\sin \left(\frac{\pi}{N} \right) \cos \left(\frac{\pi\delta}{N} \right)} \\ &= \cot \left(\frac{\pi}{N} \right) \tan \left(\frac{\pi\delta}{N} \right). \end{aligned} \quad (18)$$

In other words, an analytical estimator can be written as

$$\hat{\delta}_{qui1}^a = \frac{N}{\pi} \tan^{-1} \left\{ \tan \left(\frac{\pi}{N} \right) \frac{\operatorname{Re} \left\{ \frac{S[k_p - 1]}{S[k_p]} \right\}}{1 - \operatorname{Re} \left\{ \frac{S[k_p - 1]}{S[k_p]} \right\}} \right\}. \quad (19)$$

Similarly, the other analytical estimator is

$$\hat{\delta}_{qui2}^a = \frac{N}{\pi} \tan^{-1} \left\{ \tan \left(\frac{\pi}{N} \right) \frac{\operatorname{Re} \left\{ \frac{S[k_p + 1]}{S[k_p]} \right\}}{\operatorname{Re} \left\{ \frac{S[k_p + 1]}{S[k_p]} \right\} - 1} \right\}. \quad (20)$$

If the above two equations and Eq. (12) for Bertocco estimator are linearized, the nonlinear functions can be approximated by $\tan(x) = x$, $\tan^{-1}(x) = x$, and $\sin^{-1}(x) = x$. Then, Aboutanios has shown that we can view Quinn estimator as a linearized version of Bertocco estimator [27].

As described earlier, we can use either the criteria provided by Quinn or that of Bertocco et al. to determine which of the estimators to be used.

2.3. Macleod estimator

The estimator proposed by MacLeod [5] works as follows:

$$d_{mac} = \frac{\operatorname{Re} \{ S[k_p - 1]S[k_p]^* - S[k_p + 1]S[k_p]^* \}}{\operatorname{Re} \{ 2S[k_p]S[k_p]^* + S[k_p - 1]S[k_p]^* + S[k_p + 1]S[k_p]^* \}}, \quad (21)$$

$$\hat{\delta}_{mac} = \frac{\sqrt{1 + 8d_{mac}^2} - 1}{4d_{mac}}. \quad (22)$$

Again, applying (9), we can derive the following equation for d_{mac} :

$$\begin{aligned} &\frac{\operatorname{Re} \{ S[k_p - 1]S[k_p]^* - S[k_p + 1]S[k_p]^* \}}{\operatorname{Re} \{ 2S[k_p]S[k_p]^* + S[k_p - 1]S[k_p]^* + S[k_p + 1]S[k_p]^* \}} \\ &= \frac{\frac{\cos(\pi/N)}{\sin(\pi(\delta+1)/N)} - \frac{\cos(\pi/N)}{\sin(\pi(\delta-1)/N)}}{\frac{2}{\sin(\pi\delta/N)} + \frac{\cos(\pi/N)}{\sin(\pi(\delta+1)/N)} + \frac{\cos(\pi/N)}{\sin(\pi(\delta-1)/N)}} \\ &= \frac{\sin \left(\frac{\pi(\delta-1)}{N} \right) - \sin \left(\frac{\pi(\delta+1)}{N} \right)}{\frac{2 \sin(\pi(\delta-1)/N) \sin(\pi(\delta+1)/N)}{\cos(\pi/N) \sin(\pi\delta/N)} + \sin \left(\frac{\pi(\delta-1)}{N} \right) + \sin \left(\frac{\pi(\delta+1)}{N} \right)} \end{aligned}$$

$$\begin{aligned}
&= \frac{-2 \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right)}{2 \left[\sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) - \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right) \right] + 2 \sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi}{N}\right)} \\
&= \frac{-\sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right)}{2 \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) - \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right)}. \quad (23)
\end{aligned}$$

For the above and the following derivations, we use the following relationships: $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin(\alpha) \cos(\beta)$, $\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2 \cos(\alpha) \sin(\beta)$, and $\sin(\alpha + \beta) \cdot \sin(\alpha - \beta) = \sin^2(\alpha) \cos^2(\beta) - \cos^2(\alpha) \sin^2(\beta)$.

Then, $\sqrt{1 + 8d_{mac}^2/4d_{mac}}$ can be derived as

$$\begin{aligned}
\frac{\sqrt{1 + 8d_{mac}^2}}{4d_{mac}} &= \sqrt{\frac{1}{16d_{mac}^2} + \frac{1}{2}} \\
&= \sqrt{\frac{4 \sin^4\left(\frac{\pi\delta}{N}\right) \cos^4\left(\frac{\pi}{N}\right) + \cos^4\left(\frac{\pi\delta}{N}\right) \sin^4\left(\frac{\pi}{N}\right) - 4P}{16P} + \frac{1}{2}} \\
&\quad \left(\text{Note : } P = \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right) \cos^2\left(\frac{\pi}{N}\right) \right) \\
&= \sqrt{\frac{\left[2 \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) + \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right) \right]^2}{\left[4 \sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right) \right]^2}} \\
&= \frac{2 \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) + \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right)}{4 \sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right)}. \quad (24)
\end{aligned}$$

Finally, $\left(\sqrt{1 + 8d_{mac}^2} - 1\right)/4d_{mac}$ can be derived from Eqs. (23) and (24) as

$$\begin{aligned}
&\frac{\sqrt{1 + 8d_{mac}^2}}{4d_{mac}} - \frac{1}{4d_{mac}} \\
&= \frac{2 \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) + \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right)}{4 \sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right)} \\
&\quad + \frac{2 \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) - \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right)}{4 \sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right)} \\
&= \frac{4 \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right)}{4 \sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right)} \\
&= \cot\left(\frac{\pi}{N}\right) \tan\left(\frac{\pi\delta}{N}\right). \quad (25)
\end{aligned}$$

Therefore, the analytical solution for MacLeod estimator is

$$\hat{\delta}_{mac}^a = \frac{N}{\pi} \tan^{-1} \left\{ \tan\left(\frac{\pi}{N}\right) \left[\frac{\sqrt{1 + 8d_{mac}^2} - 1}{4d_{mac}} \right] \right\}. \quad (26)$$

2.4. Jacobsen and Candan estimators

The estimator proposed by Jacobsen et al. [7] works as follows:

$$\hat{\delta}_{jac} = \text{Re} \left\{ \frac{(S[k_p - 1] - S[k_p + 1])}{2S[k_p] - S[k_p - 1] - S[k_p + 1]} \right\}. \quad (27)$$

Candan [8] later improved it by multiplying a scaling factor:

$$\hat{\delta}_{can} = \frac{\tan(\pi/N)}{\pi/N} \hat{\delta}_{jac}. \quad (28)$$

The reason for the scaling factor will be apparent after the analysis.

Applying (9) and some of the results from (23), we can derive the analytical expression in the following equation:

$$\begin{aligned}
&\text{Re} \left\{ \frac{(S[k_p - 1] - S[k_p + 1])}{2S[k_p] - S[k_p - 1] - S[k_p + 1]} \right\} \\
&= \text{Re} \left\{ \frac{\frac{e^{-j\pi/N}}{\sin(\pi(\delta+1)/N)} - \frac{e^{j\pi/N}}{\sin(\pi(\delta-1)/N)}}{\frac{2}{\sin(\pi\delta/N)} - \frac{e^{-j\pi/N}}{\sin(\pi(\delta+1)/N)} - \frac{e^{j\pi/N}}{\sin(\pi(\delta-1)/N)}} \right\} \\
&= \text{Re} \left\{ \frac{-2 \cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right) - j2 \sin\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \cos\left(\frac{\pi}{N}\right)}{\frac{2Q}{\sin(\pi\delta/N)} - 2 \sin\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) - j2 \cos\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right)} \right\} \\
&\quad \left(\text{Note : } Q = \sin^2\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{N}\right) - \cos^2\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{N}\right) \right) \\
&= \text{Re} \left\{ \frac{\left[\frac{\sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi}{N}\right)}{\cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right)} \left[\cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) + j \sin\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \right] \right]}{\left[\cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \left[\cos\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) + j \sin\left(\frac{\pi\delta}{N}\right) \sin\left(\frac{\pi}{N}\right) \right] \right]} \right\} \\
&= \cot\left(\frac{\pi}{N}\right) \tan\left(\frac{\pi\delta}{N}\right). \quad (29)
\end{aligned}$$

The same result has been derived by Candan [12].

Thus, the analytical solution for the Jacobsen estimator is

$$\hat{\delta}_{jac}^a = \frac{N}{\pi} \tan^{-1} \left\{ \tan\left(\frac{\pi}{N}\right) \text{Re} \left\{ \frac{(S[k_p - 1] - S[k_p + 1])}{2S[k_p] - S[k_p - 1] - S[k_p + 1]} \right\} \right\}. \quad (30)$$

The reason for Candan's scaling factor is also apparent from (29). Its first order approximation is

$$\cot\left(\frac{\pi}{N}\right) \tan\left(\frac{\pi\delta}{N}\right) \approx \cot\left(\frac{\pi}{N}\right) \frac{\pi}{N} \delta. \quad (31)$$

The scaling factor removes the $\cot(\pi/N) \cdot \pi/N$ term from the equation and, thereby, improves the approximation. In fact, the same scaling factor can be applied to both Quinn and MacLeod estimators because they have similar analytical expressions.

2.5. Aboutanios estimator

The estimator proposed by Aboutanios and Mulgrew [17] is an iterative estimator. It is considered the best iterative estimator in terms of noise performance because it can approach CRLB asymptotically in two iterations. There are two versions for this estimator. In the first

iteration, the first version is

$$\hat{\delta}_{abo1} = \frac{1}{2} \operatorname{Re} \left\{ \frac{S \left[k_p + \frac{1}{2} \right] + S \left[k_p - \frac{1}{2} \right]}{S \left[k_p + \frac{1}{2} \right] - S \left[k_p - \frac{1}{2} \right]} \right\} \quad (32)$$

and the second version is

$$\hat{\delta}_{abo2} = \frac{1}{2} \cdot \frac{|S \left[k_p + \frac{1}{2} \right]| - |S \left[k_p - \frac{1}{2} \right]|}{|S \left[k_p + \frac{1}{2} \right]| + |S \left[k_p - \frac{1}{2} \right]|}. \quad (33)$$

For the i -th iteration, two additional shifted DFT coefficients need to be calculated

$$X_d = \sum_{n=0}^{N-1} s[n] e^{-j2\pi n(k_p + \hat{\delta}_{i-1}^{abo} + d)/N}, \quad d = \pm 0.5 \quad (34)$$

where $\hat{\delta}_i^{abo}$ is the estimated frequency offset at the i -th iteration and it is calculated as

$$h(\hat{\delta}_{i-1}) = \begin{cases} \frac{1}{2} \operatorname{Re} \left\{ \frac{X_{0.5} + X_{-0.5}}{X_{0.5} - X_{-0.5}} \right\} & \text{for version 1} \\ \frac{1}{2} \frac{|X_{0.5}| - |X_{-0.5}|}{|X_{0.5}| + |X_{-0.5}|} & \text{for version 2} \end{cases} \quad (35)$$

$$\hat{\delta}_i^{abo} = \hat{\delta}_{i-1}^{abo} + h(\hat{\delta}_{i-1}). \quad (36)$$

From version 1 of the above estimator, Aboutanios [27] has shown that a variation has the following analytical solution:

$$\hat{\delta}_{abo}^v = \frac{N}{2\pi} \arg \left\{ \frac{S \left[k_p + \frac{1}{2} \right] - S \left[k_p - \frac{1}{2} \right]}{e^{-j(\pi/N)} S \left[k_p + \frac{1}{2} \right] - e^{j(\pi/N)} S \left[k_p - \frac{1}{2} \right]} \right\}. \quad (37)$$

The following discussion will focus on the iterative Aboutanios estimator. To prevent redundancy, when we refer to Aboutanios estimator from now on, it means the iterative estimator but not the analytical estimator in (37) and we will show that both versions of the iterative estimator also have analytical solutions.

For the first version of the iterative estimator, it can be derived by applying (9) as follows:

$$\begin{aligned} & \frac{1}{2} \operatorname{Re} \left\{ \frac{S \left[k_p + \frac{1}{2} \right] + S \left[k_p - \frac{1}{2} \right]}{S \left[k_p + \frac{1}{2} \right] - S \left[k_p - \frac{1}{2} \right]} \right\} \\ &= \frac{1}{2} \operatorname{Re} \left\{ \frac{j e^{j(\pi/2N)} \frac{\cos(\pi\delta)}{\sin\left(\frac{\pi(\delta-1/2)}{N}\right)} + j e^{-j(\pi/2N)} \frac{\cos(\pi\delta)}{\sin\left(\frac{\pi(\delta+1/2)}{N}\right)}}{j e^{j(\pi/2N)} \frac{\cos(\pi\delta)}{\sin\left(\frac{\pi(\delta-1/2)}{N}\right)} - j e^{-j(\pi/2N)} \frac{\cos(\pi\delta)}{\sin\left(\frac{\pi(\delta+1/2)}{N}\right)}} \right\} \\ &= \frac{1}{2} \operatorname{Re} \left\{ \frac{e^{j(\pi/2N)} \sin\left(\frac{\pi(\delta+1/2)}{N}\right) + e^{-j(\pi/2N)} \sin\left(\frac{\pi(\delta-1/2)}{N}\right)}{e^{j(\pi/2N)} \sin\left(\frac{\pi(\delta+1/2)}{N}\right) - e^{-j(\pi/2N)} \sin\left(\frac{\pi(\delta-1/2)}{N}\right)} \right\} \\ &= \frac{1}{2} \operatorname{Re} \left\{ \frac{\sin\left(\frac{\pi\delta}{N}\right) \cos^2\left(\frac{\pi}{2N}\right) + j \cos\left(\frac{\pi\delta}{N}\right) \sin^2\left(\frac{\pi}{2N}\right)}{\sin\left(\frac{\pi}{2N}\right) \cos\left(\frac{\pi}{2N}\right) \left[\cos\left(\frac{\pi\delta}{N}\right) + j \sin\left(\frac{\pi\delta}{N}\right) \right]} \right\} \end{aligned}$$

$$= \frac{\sin\left(\frac{\pi\delta}{N}\right) \cos\left(\frac{\pi\delta}{N}\right) \left[\cos^2\left(\frac{\pi}{2N}\right) + \sin^2\left(\frac{\pi}{2N}\right) \right]}{2 \sin\left(\frac{\pi}{2N}\right) \cos\left(\frac{\pi}{2N}\right) \left[\cos^2\left(\frac{\pi\delta}{N}\right) + \sin^2\left(\frac{\pi\delta}{N}\right) \right]} = \frac{\sin\left(\frac{2\pi\delta}{N}\right)}{2 \sin\left(\frac{\pi}{N}\right)}. \quad (38)$$

Therefore, the analytical solution for version 1 of Aboutanios estimator is

$$\hat{\delta}_{abo1}^a = \frac{N}{2\pi} \sin^{-1} \left\{ \sin\left(\frac{\pi}{N}\right) \operatorname{Re} \left\{ \frac{S \left[k_p + \frac{1}{2} \right] + S \left[k_p - \frac{1}{2} \right]}{S \left[k_p + \frac{1}{2} \right] - S \left[k_p - \frac{1}{2} \right]} \right\} \right\}. \quad (39)$$

For the second version, the absolute value operation can be removed by applying the fact that $|\delta| < 1/2$ to (9). The analytical expression can be derived as

$$\begin{aligned} & \frac{1}{2} \cdot \frac{|S \left[k_p + \frac{1}{2} \right]| - |S \left[k_p - \frac{1}{2} \right]|}{|S \left[k_p + \frac{1}{2} \right]| + |S \left[k_p - \frac{1}{2} \right]|} \\ &= \frac{1}{2} \cdot \frac{\sin\left(\frac{\pi(\delta+1/2)}{N}\right) + \sin\left(\frac{\pi(\delta-1/2)}{N}\right)}{\sin\left(\frac{\pi(\delta+1/2)}{N}\right) - \sin\left(\frac{\pi(\delta-1/2)}{N}\right)} \\ &= \frac{\cos\left(\frac{\pi}{2N}\right) \sin\left(\frac{\pi\delta}{N}\right)}{2 \sin\left(\frac{\pi}{2N}\right) \cos\left(\frac{\pi\delta}{N}\right)} = \frac{\tan\left(\frac{\pi\delta}{N}\right)}{2 \tan\left(\frac{\pi}{2N}\right)}. \end{aligned}$$

Therefore, the analytical solution for version 2 of Aboutanios estimator is

$$\hat{\delta}_{abo2}^a = \frac{N}{\pi} \tan^{-1} \left\{ \tan\left(\frac{\pi}{2N}\right) \frac{|S \left[k_p + \frac{1}{2} \right]| - |S \left[k_p - \frac{1}{2} \right]|}{|S \left[k_p + \frac{1}{2} \right]| + |S \left[k_p - \frac{1}{2} \right]|} \right\}. \quad (40)$$

Although Aboutanios estimator is considered best in terms of noise performance, the bias for its linearized version in the first iteration is worse than Candan and phase-corrected Quinn estimators [11]. In the i -th iteration, the bias is proportional to $1/N^{2i}$ [30] and will eventually diminish as the number of iteration increases. Since an analytical expression can achieve unbiased estimation without iteration and the main concern of this paper is on the bias, we will only compare the analytical expression against the first iteration of Aboutanios estimator.

Also, when the two versions are used as approximate estimators, they can be improved by following Candan's idea of applying scaling factor to the interpolation. For the first version, we can use a scaling factor of $\sin(\pi/N) \cdot N/\pi$. For the second version, we can use a scaling factor of $\tan(\pi/(2N)) \cdot 2N/\pi$.

2.6. Discussion

After the theoretical analysis of existing estimators, the obvious question is whether there are other estimators that have analytical solutions. To answer the question, we have to lay out the three requirements for this kind of

estimators. The first requirement is the ability to remove the common terms $A_0 \exp(j\theta_0) \exp(j(N-1)\pi\delta/N)$ from the DFT coefficients. This has to be done by dividing two different combinations of DFT coefficients. MacLeod estimator uses a slightly different approach because it applies the real operation separately to the numerator and denominator. It uses the conjugate of $S[k_p]$ to remove the common phase term prior to the real operation and, then, uses the division to remove the common amplitude term. The second requirement is the stability of the estimator. Because of the division to remove the common terms, the denominator cannot be zero in the range of δ . Otherwise, the estimator becomes unstable. Therefore, for the estimators using integer-indexed DFT coefficients, some form of $S[k_p]$ must be present in the denominator because $S[k_p]$ has the maximum magnitude and cannot be zero unless the signal is zero. For $S[k_p \pm 1/2]$, we can see from (9) that they are not zero when $|\delta| < 1/2$. Therefore, the Aboutanios estimator inherently satisfies the stability condition. The third requirement is that the distance between the DFT coefficients used in the interpolation must be regular so that the trigonometric properties can be applied to simplify the equations. For example, if we want to modify the Aboutanios estimator, the next candidates to be added to the interpolation are $S[k_p \pm 3/2]$. If we insert $S[k_p]$ into the equation, it disrupts the regularity and the $\cos(\pi\delta)$ term in the numerators of $S[k_p \pm 1/2]$ cannot be eliminated.

Given these three requirements, the choices of the parameters are quite limited. Therefore, some variations of the existing estimators are possible but an entirely new estimator is difficult to find. In the following, we will give some examples of these variations.

For Jacobsen estimator, we have the following variation:

$$d_{jac} = \text{Re} \left\{ \frac{(e^{j(\pi/N)} S[k_p - 1] - e^{-j(\pi/N)} S[k_p + 1])}{2S[k_p] - S[k_p - 1] - S[k_p + 1]} \right\},$$

$$\hat{\delta}_{jac}^v = \frac{N}{2\pi} \sin^{-1} \left(2 \sin \left(\frac{\pi}{N} \right) d_{jac} \right). \quad (41)$$

For Aboutanios estimator, we have the following variation:

$$d_{abo} = \frac{1}{2} \text{Re} \left\{ \frac{e^{-j(\pi/2N)} S \left[k_p + \frac{1}{2} \right] + e^{j(\pi/2N)} S \left[k_p - (1/2) \right]}{S \left[k_p + \frac{1}{2} \right] - S \left[k_p - \frac{1}{2} \right]} \right\},$$

$$\hat{\delta}_{abo}^v = \frac{N}{2\pi} \sin^{-1} \left(4 \sin \left(\frac{\pi}{2N} \right) d_{abo} \right). \quad (42)$$

The performance of the variations is almost identical to the original estimator. Hence, the given variations are observed to have limited innovation value.

3. Experimental results

For all experiments in this section, we used the following parameters: $A_0 = 1$, $\theta = 0$, $f_s = 1$, $N = 64$, and $k_p = 10$. First, a simple experiment is conducted to verify that the analytical results are correct. We vary δ from -0.49 to 0.49 with step size 0.01 and set the noise to zero. The error between the true offset and the estimated offset ($\delta - \hat{\delta}$) is then calculated. Since there are a large number of estimators being tested, we will call the original version

of the estimators: Quinn (Eqs. (14) and (15)), MacLeod (Eq. (22)), Jacobsen (Eq. (27)), Candan (Eq. (28)), and Aboutanios (Eqs. (32) and (33)), collectively as the “linearized” estimators.

The analytical version of the estimators: Bertocco (Eqs. (10) and (13)), Quinn (Eqs. (19) and (20)), MacLeod (Eq. (26)), Jacobsen (Eq. (30)) and Aboutanios (Eqs. (39) and (40)), is collectively called as “bias removed” estimators. For all *bias removed* estimators, the absolute values of the errors are all less than 4×10^{-15} . Testing on other parameters also yields the same results. Therefore, we can confidently say that the estimation is completely accurate when noise is not present. As a comparison, for *linearized* estimators, the maximum absolute errors for Quinn, MacLeod, and Jacobsen estimators are 2.9919×10^{-4} at $\delta = \pm 0.49$ and the maximum absolute error for Candan estimator is 9.4516×10^{-5} at $\delta = \pm 0.49$.

Although the main issue discussed in this paper is the bias, it is still worth observing the performance of the *bias removed* estimators under the influence of noise. White Gaussian noise is added to the signal while δ varies from -0.45 to 0.45 with step size 0.05 . The error between the true offset and the estimated offset is calculated as the previous experiment. Each experiment is repeated 10,000 times and MSE is calculated. Fig. 1 plots MSE vs. δ at two different SNR's. From the figure, we can see that Jacobsen and Aboutanios estimators are better at small δ . The two versions of Aboutanios estimator perform the same and they are better than Jacobsen estimator. On the other hand, Quinn, MacLeod, and Bertocco estimators are better at large δ and MacLeod estimator is the best of the three. Therefore, there is not a single *bias removed* estimator which shows the best performance throughout the entire range of δ . To obtain an optimal estimator in terms of noise interference, either a new estimator has to be found or a hybrid approach combining the two best estimators is necessary. Thus, the choice of the estimators depends on the applications. To fully understand the issue, we need a theoretical analysis of the noise behavior. In this regard, Candan has already done a complete analysis for Jacobsen and Candan estimators [12]. Similar analysis could be achieved by following the same analysis process. However, noise analysis requires linearized approximation of the interpolation formula and this deviates from the focus of this paper, i.e., the nonlinear nature of the estimation. Therefore, we will not address it in this paper and will consider it in the future.

From Fig. 1, we can also see that the behavior of each estimator follows the same trend at different SNR. To better illustrate that MSE and SNR have a linear relationship in log space, Fig. 2 shows MSE vs. SNR at $\delta = 0.25$ for the two best *bias removed* estimators: MacLeod and Aboutanios, and the *linearized* estimators. We can clearly see that the MSE's for the *linearized* estimators become flat after certain SNR because the approximation error becomes the dominant source of error. On the other hand, MSE vs. SNR is linear for the analytical estimators in the entire range of SNR.

Fig. 3 shows MSE vs. δ for the *bias removed* and *linearized* estimators at high SNR. From the figure, we can see that the *bias removed* estimators have overwhelming advantage over the *linearized* estimators at high SNR. At low SNR, the performance of the estimators is predominantly

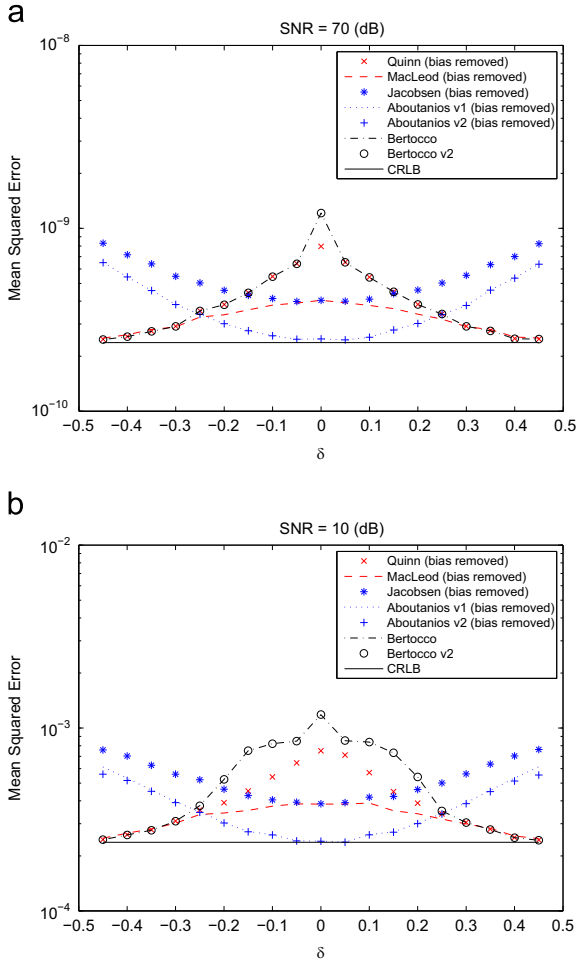


Fig. 1. Comparison of MSE vs. δ for the *bias removed* estimators at (a) high SNR (SNR = 70 dB) and (b) low SNR (SNR = 10 dB).

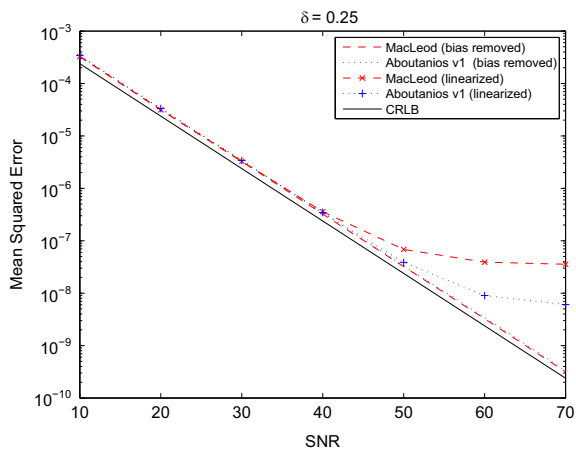


Fig. 2. Comparison of MSE vs. SNR for the *bias removed* and *linearized* estimators at $\delta = 0.25$.

determined by noise and the variation across δ is very small. Therefore, *linearized* estimators may be favorable at low SNR because it does not need nonlinear calculations. For

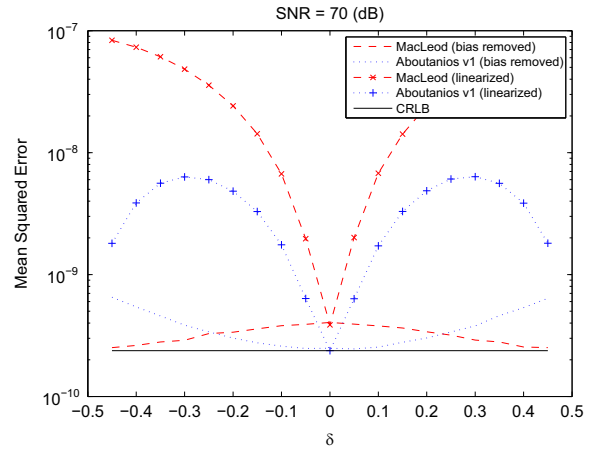


Fig. 3. Comparison of MSE vs. δ for the *bias removed* and *linearized* estimators at high SNR (SNR = 70 dB).

this reason, one may argue that the analytical expressions have little value. However, we would like to emphasize that the analytical solutions still hold important theoretical value in understanding the working of the estimators even though they may not be used practically.

4. Conclusion

In this paper, we study the interpolation of DFT coefficients to estimate the frequency of a complex sinusoidal signal. Using trigonometric properties, we derive the analytical expressions for the interpolation. The results show that many of the interpolated estimators have simple analytical expressions. These expressions have both theoretical and practical values. Theoretically, it helps illustrate the relationship between the linearized estimator and its nonlinear analytical expression. Therefore, the bias caused by the approximation can be better understood and the reason that some interpolated estimators perform better than the other can be explained. Practically, these analytical expressions can be used as a new set of unbiased estimators. We also show a number of variations of the existing estimators using the results of the theoretical analysis. In the experiments, we show that the analytical estimators are indeed much more accurate than the original interpolated estimators. However, under the influence of noise, some of the estimators are better for small frequency offset and vice versa. For future works, a theoretical analysis of the noise behavior for these estimators will help us understand how to further improve the performance under the influence of noise.

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References

- [1] A. Omidvarnia, G. Azemi, J.M. O'Toole, B. Boashash, Robust estimation of highly-varying nonlinear instantaneous frequency of mono-component signals using a lower-order complex-time distribution, *Signal Process.* 93 (2013) 3251–3260.
- [2] D. Rife, R. Boorstyn, Single tone parameter estimation from discrete-time observations, *IEEE Trans. Inf. Theory* 20 (1974) 591–598.
- [3] B.G. Quinn, Estimating frequency by interpolation using Fourier coefficients, *IEEE Trans. Signal Process.* 42 (1994) 1264–1268.
- [4] B.G. Quinn, Estimation of frequency, amplitude, and phase from the DFT of a time series, *IEEE Trans. Signal Process.* 45 (1997) 814–817.
- [5] M.D. Macleod, Fast nearly ML estimation of the parameters of real or complex single tones or resolved multiple tones, *IEEE Trans. Signal Process.* 46 (1998) 141–148.
- [6] Y.V. Zakharov, V.M. Baronkin, T.C. Tozer, DFT-based frequency estimators with narrow acquisition range, *IEE Proc. Commun.* 148 (2001) 1–7.
- [7] E. Jacobsen, P. Kootsookos, Fast accurate frequency estimators, *IEEE Signal Process. Mag.* 24 (2007) 123–125.
- [8] C. Candan, A method for fine resolution frequency estimation from three DFT samples, *IEEE Signal Process. Lett.* 18 (2011) 351–354.
- [9] C. Yang, G. Wei, A noniterative frequency estimator with rational combination of three spectrum lines, *IEEE Trans. Signal Process.* 59 (2011) 5065–5070.
- [10] G. Wei, C. Yang, F.-J. Chen, Closed-form frequency estimator based on narrow-band approximation under noisy environment, *Signal Process.* 91 (2011) 841–851.
- [11] J.-R. Liao, C.-M. Chen, Phase correction of discrete Fourier transform coefficients to reduce frequency estimation bias of single tone complex sinusoid, *Signal Process.* 94 (2014) 108–117.
- [12] C. Candan, Analysis and further improvement of fine resolution frequency estimation method from three DFT samples, *IEEE Signal Process. Lett.* 20 (2013) 913–916.
- [13] T. Abatzoglou, A fast maximum likelihood algorithm for frequency estimation of a sinusoid based on Newton's method, *IEEE Trans. Acoust. Speech Signal Process.* 33 (1985) 77–89.
- [14] Y.V. Zakharov, T.C. Tozer, Frequency estimator with dichotomous search of periodogram peak, *Electron. Lett.* 35 (1999) 1608–1609.
- [15] S. Reisenfeld, E. Aboutanios, A new algorithm for the estimation of the frequency of a complex exponential in additive Gaussian noise, *IEEE Commun. Lett.* 7 (2003) 549–551.
- [16] E. Aboutanios, A modified dichotomous search frequency estimator, *IEEE Signal Process. Lett.* 11 (2004) 186–188.
- [17] E. Aboutanios, B. Mulgrew, Iterative frequency estimation by interpolation on Fourier coefficients, *IEEE Trans. Signal Process.* 53 (2005) 1237–1242.
- [18] I. Djurovic, Estimation of the sinusoidal signal frequency based on the marginal median DFT, *IEEE Trans. Signal Process.* 55 (2007) 2043–2051.
- [19] I. Djurovic, V.V. Lukin, Estimation of single-tone signal frequency by using the L-DFT, *Signal Process.* 87 (2007) 1537–1544.
- [20] S. Minhas, E. Aboutanios, Estimation of the frequency of a complex exponential, *Proc. IEEE ICASSP, Las Vegas, NV, March 2008*, pp. 3693–3696.
- [21] Y. Liu, Z. Nie, Z. Zhao, Q.H. Liu, Generalization of iterative Fourier interpolation algorithms for single frequency estimation, *Digital Signal Process.* 21 (2011) 141–149.
- [22] S. Djukanovic, I. Djurovic, Robust M-periodogram with dichotomous search, *Signal Process.* 91 (2011) 2410–2414.
- [23] B.G. Quinn, Recent advances in rapid frequency estimation, *Digital Signal Process.* 19 (2009) 942–948.
- [24] E. Aboutanios, Generalised DFT-based estimators of the frequency of a complex exponential in noise, in: *Proceedings of the Image Signal Processing (CISP)*, Yantai, China, October 2010, pp. 2998–3002.
- [25] J.-R. Liao, S. Lo, Analysis of estimation bias for direct frequency estimators of complex sinusoid, in: *Proceedings of the International Conference on Signal-Image Technology and Internet-Based Systems (SITIS)*, Kyoto, Japan, December 2013, pp. 85–92.
- [26] M. Bertocco, C. Offelli, D. Petri, Analysis of damped sinusoidal signals via a frequency-domain interpolation algorithm, *IEEE Trans. Instrum. Meas.* 43 (1994) 245–250.
- [27] E. Aboutanios, Estimation of the frequency and decay factor of a decaying exponential in noise, *IEEE Trans. Signal Process.* 58 (2010) 501–509.
- [28] E. Aboutanios, Estimating the parameters of sinusoids and decaying sinusoids in noise, *IEEE Instrum. Meas. Mag.* 14 (2011) 8–14.
- [29] F. Qian, S. Leung, Y. Zhu, W. Wong, D. Pao, W. Lau, Damped sinusoidal signals parameter estimation in frequency domain, *Signal Process.* 92 (2012) 381–391.
- [30] J.-R. Liao, C.-M. Chen, Analysis and reduction of estimation bias for an iterative frequency estimator of complex sinusoid, in: *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Vancouver, Canada, May 2013, pp. 6138–6142.